WORKED SOLUTION

# STEP III Specimen paper, Question 13

Mechanics • Rolling Hoop

**STEP III Specimen paper, Question 13**

Read the question before you go on. It is in the STEP Support Programme's question database:

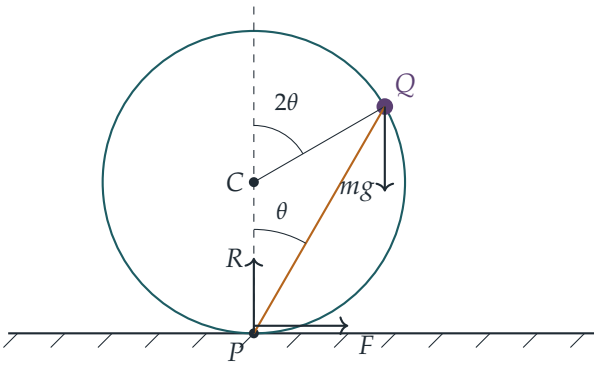
step.maths.org/questions/spec-s3-q13

## The key idea

Because the hoop is light, the only kinetic energy in the system is the particle's, and because the hoop rolls, friction does no work. So energy conservation gives the speed at once. Rolling also fixes the geometry completely: once the hoop has turned through $2\theta$, its centre has moved a distance $2a\theta$, which gives $x$ and $y$ in terms of $\theta$ alone. Finally, a light hoop can carry no net force, so the table's reaction is simply what the particle's acceleration demands, and it cannot be negative.

## Solution

Model the hoop as a light rigid circle rolling without slipping, and the particle as a point mass fixed to the rim. Let $P$ be the point of contact with the table, $C$ the centre and $Q$ the particle. Take the hoop to roll in the positive $x$ direction, so that $\theta$ increases from 0.




Here $R$ and $F$ are the normal reaction and friction of the table on the hoop at $P$, and the chord $PQ$ makes angle $\theta$ with the vertical (half the angle $2\theta$ at the centre).

### The speed

Both $R$ and $F$ act at $P$, which is instantaneously at rest because the hoop rolls, so neither does any work. The hoop is light, so the only kinetic energy is the particle's and the only potential energy change is the particle's. The particle starts at height $2a$, essentially at rest, and when the radius $CQ$ makes angle $2\theta$ with the upward vertical its height is $a + a\cos 2\theta = 2a\cos^2\theta$. Energy conservation gives

$$\tfrac{1}{2}mv^2 = mg\big(2a - 2a\cos^2\theta\big) = 2mga\sin^2\theta,$$

so $v^2 = 4ga\sin^2\theta$ and, since $0 \le \theta < \frac{\pi}{2}$ gives $\sin\theta \ge 0$,

$$v = 2(ga)^{1/2}\sin\theta,$$

as required.

### The position and $\dot\theta$

The hoop has turned through angle $2\theta$, and rolling without slipping means the centre has moved forward by the arc length $a \cdot 2\theta$. Relative to $C$, the particle is $a\sin 2\theta$ ahead and $a\cos 2\theta$ above. So

$$x = 2a\theta + a\sin 2\theta, \qquad y = a + a\cos 2\theta = 2a\cos^2\theta.$$

Differentiating with respect to time,

$$\dot x = 2a\dot\theta\,(1 + \cos 2\theta) = 4a\dot\theta\cos^2\theta,$$
$$\dot y = -2a\dot\theta\sin 2\theta = -4a\dot\theta\sin\theta\cos\theta,$$

and therefore

$$v^2 = \dot x^2 + \dot y^2 = 16a^2\dot\theta^2\cos^2\theta\big(\cos^2\theta + \sin^2\theta\big) = 16a^2\dot\theta^2\cos^2\theta.$$

? **Ask yourself** Is there a quicker way to see that $v = 4a\dot\theta\cos\theta$?

Yes: $P$ is the instantaneous centre of rotation, and the hoop turns at rate $2\dot\theta$. The chord $PQ$ subtends the angle $\pi - 2\theta$ at $C$, so $PQ = 2a\cos\theta$, and $v = PQ \cdot 2\dot\theta = 4a\dot\theta\cos\theta$, matching the calculation.

With $\dot\theta > 0$ and $\cos\theta > 0$, comparing with the speed found from energy,

$$4a\dot\theta\cos\theta = 2(ga)^{1/2}\sin\theta \quad\Longrightarrow\quad \dot\theta = \tfrac{1}{2}(g/a)^{1/2}\tan\theta,$$

which is the first required result. (The left-hand side must be the rate $\dot\theta$, not $\theta$: the right-hand side has units of $\mathrm{s}^{-1}$.)

### The vertical acceleration

Substituting $\dot\theta$ into $\dot y$,

$$\dot y = -4a\sin\theta\cos\theta\cdot\tfrac{1}{2}(g/a)^{1/2}\frac{\sin\theta}{\cos\theta} = -2(ga)^{1/2}\sin^2\theta.$$

Differentiating again and using $\dot\theta$ once more,

$$\ddot y = -4(ga)^{1/2}\sin\theta\cos\theta\,\dot\theta = -4(ga)^{1/2}\sin\theta\cos\theta\cdot\tfrac12(g/a)^{1/2}\frac{\sin\theta}{\cos\theta} = -2g\sin^2\theta,$$

as required. In the same way $\dot x = 4a\cos^2\theta\,\dot\theta = (ga)^{1/2}\sin 2\theta$, and

$$\ddot x = 2(ga)^{1/2}\cos 2\theta\,\dot\theta = g\cos 2\theta\tan\theta.$$

#### What stops the rolling

The hoop is light, so the total force on the hoop-and-particle system equals $m$ times the particle's acceleration. The external forces are $mg$ down, $R$ up and $F$ horizontally. Resolving vertically,

$$R - mg = m\ddot y \quad\Longrightarrow\quad R = mg\big(1-2\sin^2\theta\big) = mg\cos 2\theta,$$

and horizontally $F = m\ddot x = mg\cos 2\theta\tan\theta$, so that $F/R = \tan\theta$ while $R > 0$.

**Ask yourself** What goes wrong when $\theta$ passes $\pi/4$?

The table can push but not pull, so $R \geq 0$. But $R = mg\cos 2\theta$ falls to zero at $\theta = \pi/4$, when the particle is level with the centre, and rolling beyond that would need $R < 0$. Equivalently, continued rolling would need $\ddot y = -2g\sin^2\theta < -g$: the particle would have to accelerate downwards faster than free fall, which gravity alone cannot provide. So at $\theta = \pi/4$ the contact force vanishes and the hoop leaves the table. The particle then moves as a projectile, starting with velocity components $(ga)^{1/2}$ forwards and $(ga)^{1/2}$ downwards, and the light hoop is simply carried along with it, its lowest point rising off the table.

Friction may intervene earlier. Rolling needs $|F| \leq \mu R$, that is $\tan\theta \leq \mu$, and $\tan\theta$ increases to 1 as $\theta \to \pi/4$. So if the coefficient of friction satisfies $\mu < 1$, the hoop starts to slip when $\tan\theta = \mu$, before reaching $\theta = \pi/4$. Only if $\mu \geq 1$ does it roll all the way to $\theta = \pi/4$ and then jump.

## Checks

- At $\theta = 0$, $\dot\theta = 0$: the top position is an equilibrium, which is why the hoop has to be displaced slightly.
- At $\theta = \pi/4$: $v = (2ga)^{1/2}$, which is the speed from a fall of $a$, and indeed the particle has dropped from $2a$ to $a$.
- The computer integrated the equation of motion for $\phi = 2\theta$ independently (from the Lagrangian $ma^2(1+\cos\phi)\dot\phi^2 - mga(1+\cos\phi)$) and compared finite-difference velocities and accelerations with the formulae. The speed, $\dot\theta$, $\ddot y$, $R = mg\cos 2\theta$ and $F/R = \tan\theta$ all agreed to within $10^{-6}$; for example at $\theta = \pi/4$ it gives $R = 0$ and $F/R = 1$.

## Common slips

- Taking the centre to move a distance $a\theta$ rather than $2a\theta$: the hoop has turned through $2\theta$, not $\theta$.
- Including friction in the energy equation. The contact point is at rest, so friction does no work while the hoop rolls.
- Saying only that the particle "falls off". The particle is attached; it is the hoop that leaves the table, because the reaction would have to become negative.

## Beyond STEP

The contact point of a rolling body is its instantaneous centre of rotation, and rolling without slipping is a constraint that removes a degree of freedom without doing work. Such constraints are handled systematically by Lagrangian mechanics, where one writes the energy in a single coordinate such as $\theta$ and the equation of motion follows automatically; the check above did exactly this. Questions of when a constraint force vanishes (a bead leaving a sphere, a car leaving a hump) are typical of first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rolling motion and constraint forces in detail.

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